

Refresher Course in Calculus, Probability, and Statistics

Day 5b: Matrix Operations

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Linear Algebra

- ▶ **Matrix:** a rectangular array (collection) of numbers.
- ▶ An $m \times n$ (read: m -by- n) matrix has m rows and n columns:

$$\mathbf{A}_{m \times n} = (a_{ij})_{m \times n} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

where $(a_{ij})_{m \times n}$ denotes the element a in row i and column j .
If:

- ▶ $m = 1$ and $n > 1$: **row vector**;
- ▶ $m > 1$ and $n = 1$: **column vector**;
- ▶ $m = n$: **square matrix**;
- ▶ $m = n = 1$: **scalar**.



James Joseph
Sylvester
1814–1897

Matrix Operations: equality of matrices

- ▶ Let $\mathbf{A}_{m \times n} = (a_{ij})_{m \times n}$, $\mathbf{B}_{p \times q} = (b_{ij})_{p \times q}$, $\alpha_{1 \times 1}$.
- ▶ $\mathbf{A}$ and $\mathbf{B}$ are **equal** iff:
 - same order (same dimensions): $m = p$ and $n = q$;
 - identical elements in corresponding locations: $a_{ij} = b_{ij}$ for all i, j .
- ▶ Examples:

$$\begin{pmatrix} 4 & 2 \\ 5 & 0 \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 5 & 0 \end{pmatrix} \neq \begin{pmatrix} 5 & 0 \\ 4 & 2 \end{pmatrix} \neq \begin{pmatrix} 4 & 2 & 1 \\ 5 & 0 & -5 \end{pmatrix}.$$

Matrix Operations: addition and subtraction

- ▶ Matrices can be added or subtracted only if they have the **same order** (conformability condition for addition).
- ▶ **Addition** — sum each pair of corresponding elements: $\mathbf{A} + \mathbf{B} = (a_{ij} + b_{ij})$.

$$\begin{pmatrix} 4 & 2 \\ 5 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 2 \\ 2 & -3 \end{pmatrix} = \begin{pmatrix} 4+1 & 2+2 \\ 5+2 & 0+(-3) \end{pmatrix} = \begin{pmatrix} 5 & 4 \\ 7 & -3 \end{pmatrix}.$$

- ▶ **Subtraction** is similar: $\mathbf{A} - \mathbf{B} = (a_{ij} - b_{ij})$.

$$\begin{pmatrix} 4 & 2 \\ 5 & 0 \end{pmatrix} - \begin{pmatrix} 1 & 2 \\ 2 & -3 \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 3 & 3 \end{pmatrix}.$$

Matrix Operations: scalar multiplication

- ▶ Multiplication by a scalar α : multiply each element by α , $\alpha \mathbf{A}_{m \times n} = (\alpha a_{ij})_{m \times n}$.
- One can invert scalar multiplication by extracting a common factor.
- ▶ Examples:

$$3 \begin{pmatrix} 4 & 2 \\ 5 & 0 \end{pmatrix} = \begin{pmatrix} 12 & 6 \\ 15 & 0 \end{pmatrix}, \quad \begin{pmatrix} -2 & -1 \\ -\frac{5}{2} & 0 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} 4 & 2 \\ 5 & 0 \end{pmatrix}.$$

Matrix Operations: multiplication

- ▶ **A** and **B** can be multiplied only if the column dimension of **A** equals the row dimension of **B** (conformability for multiplication):
 - with $\mathbf{A}_{m \times n}$ and $\mathbf{B}_{p \times q}$, $\mathbf{AB}$ is defined iff $n = p$;
 - the product $\mathbf{C} = \mathbf{AB}$ has as many rows as **A** and as many columns as **B**:
$$\mathbf{A}_{m \times n} \mathbf{B}_{p \times q} = \mathbf{C}_{m \times q}.$$
- ▶ c_{ij} is built from the i th row of **A** and the j th column of **B**: pair the elements, multiply each pair, and sum the products:

$$c_{ij} = \sum_{r=1}^n a_{ir}b_{rj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}.$$

- ▶ *Note:* there is no matrix division.

Matrix Operations: multiplication (continued)

Example 1: $\mathbf{A}_{2 \times 3} \mathbf{B}_{3 \times 2} = \mathbf{C}_{2 \times 2}$

$$\begin{pmatrix} 1 & 0 & 3 \\ 2 & 1 & 5 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 2 & 5 \\ 6 & 2 \end{pmatrix} = \begin{pmatrix} 1 \cdot 1 + 0 \cdot 2 + 3 \cdot 6 & 1 \cdot 3 + 0 \cdot 5 + 3 \cdot 2 \\ 2 \cdot 1 + 1 \cdot 2 + 5 \cdot 6 & 2 \cdot 3 + 1 \cdot 5 + 5 \cdot 2 \end{pmatrix} = \begin{pmatrix} 19 & 9 \\ 34 & 21 \end{pmatrix}$$

Example 2: $\mathbf{A}_{3 \times 2} \mathbf{B}_{2 \times 1} = \mathbf{C}_{3 \times 1}$

$$\begin{pmatrix} 1 & 3 \\ -2 & 8 \\ 0 & 12 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \cdot 2 + 3 \cdot 1 \\ -2 \cdot 2 + 8 \cdot 1 \\ 0 \cdot 2 + 12 \cdot 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 4 \\ 12 \end{pmatrix}$$

Matrix Operations: properties

► Matrix addition

- commutative: $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$;
- associative: $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$.

► Matrix multiplication

- *not* commutative: $\mathbf{AB} \neq \mathbf{BA}$ (with exceptions);
- associative: $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$;
- distributive (pre-multiplication): $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$;
- distributive (post-multiplication): $(\mathbf{B} + \mathbf{C})\mathbf{A} = \mathbf{BA} + \mathbf{CA}$.

► Idiosyncrasies of matrix algebra

- there is no matrix division;
- $\mathbf{AB} = \mathbf{0}$ does *not* imply $\mathbf{A} = \mathbf{0}$ or $\mathbf{B} = \mathbf{0}$ (can occur when $\mathbf{A}, \mathbf{B}$ are singular);
- $\mathbf{AB} = \mathbf{AC}$ with $\mathbf{A} \neq \mathbf{0}$ does *not* imply $\mathbf{B} = \mathbf{C}$ (can occur when the matrices are singular, see below).

Matrix Operations: powers

- For a square matrix $\mathbf{A}_{n \times n}$:

$$\mathbf{A}^2 = \mathbf{A}\mathbf{A}, \quad \mathbf{A}^3 = \mathbf{A}\mathbf{A}\mathbf{A}, \quad \dots, \quad \mathbf{A}^N = \underbrace{\mathbf{A}\mathbf{A} \cdots \mathbf{A}}_{N \text{ times}}.$$

► Example:
$$\begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}^2 = \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix} = \begin{pmatrix} 9 & 8 \\ 16 & 17 \end{pmatrix}.$$

► **Idempotent matrix** ($\mathbf{A}^2 = \mathbf{A}$):
$$\begin{pmatrix} 2 & 1 \\ -2 & -1 \end{pmatrix}^2 = \begin{pmatrix} 2 & 1 \\ -2 & -1 \end{pmatrix}.$$

- **Diagonal matrix** — powers act element-wise: if $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$ then

$$\mathbf{D}^m = \text{diag}(d_1^m, \dots, d_n^m); \text{ e.g. } \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}.$$

Matrix Operations: the identity matrix

- **Identity matrix:** a square matrix with ones on its main diagonal and zeros elsewhere:

$$\mathbf{I}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{I}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{I}_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

- Same role as 1 in scalar algebra:
 - $\mathbf{A}_{m \times n} \mathbf{I}_n = \mathbf{I}_m \mathbf{A}_{m \times n} = \mathbf{A}_{m \times n}$;
 - you can insert or delete an identity without changing a product:
 $\mathbf{A}_{m \times n} \mathbf{I}_{n \times n} \mathbf{B}_{n \times p} = \mathbf{A}_{m \times n} \mathbf{B}_{n \times p}$;
 - $\mathbf{I}_n^k = \mathbf{I}_n$ (idempotent).

Matrix Operations: the null matrix

- **Null (zero) matrix:** a matrix with zeros everywhere; same role as 0 in scalar algebra:

$$\mathbf{0}_{2 \times 2} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{0}_{2 \times 3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

- Properties:

- $\mathbf{A}_{m \times n} + \mathbf{0}_{m \times n} = \mathbf{0}_{m \times n} + \mathbf{A}_{m \times n} = \mathbf{A}_{m \times n};$
- $\mathbf{A}_{m \times n} \mathbf{0}_{n \times p} = \mathbf{0}_{m \times p}$ and $\mathbf{0}_{q \times m} \mathbf{A}_{m \times n} = \mathbf{0}_{q \times n}.$

Matrix Operations: transpose

- ▶ Interchanging the rows and columns of a matrix $\mathbf{A}$ yields its **transpose**, denoted $\mathbf{A}'$ or $\mathbf{A}^T$:

$$\mathbf{A}_{2 \times 3} = \begin{pmatrix} 3 & 8 & -2 \\ 0 & 1 & 9 \end{pmatrix} \implies \mathbf{A}'_{3 \times 2} = \begin{pmatrix} 3 & 0 \\ 8 & 1 \\ -2 & 9 \end{pmatrix}$$

(reflection of the elements along the main diagonal).

- ▶ Properties:
 - $(\mathbf{A}')' = \mathbf{A}$ (transpose is an involution);
 - $(\mathbf{A} + \mathbf{B})' = \mathbf{A}' + \mathbf{B}'$ (transpose respects addition);
 - $(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'$ (transpose of a product is the product of transposes in *reverse* order);
 - $(\alpha\mathbf{A})' = \alpha\mathbf{A}'$ (transpose of a scalar is the same scalar).
- ▶ If $\mathbf{A}'_{n \times n} = \mathbf{A}_{n \times n}$, the matrix is **symmetric**:

$$\mathbf{D} = \begin{pmatrix} 1 & 2 & -1 \\ 2 & -4 & 3 \\ -1 & 3 & 9 \end{pmatrix} = \mathbf{D}'.$$

Matrix Operations: determinant

- ▶ The **determinant** of a square matrix $\mathbf{A}$, denoted $|\mathbf{A}|$ or $\det \mathbf{A}$, is a scalar derived from $\mathbf{A}$.
- ▶ 2×2 matrix:

$$\mathbf{A}_{2 \times 2} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \quad |\mathbf{A}| = a_{11}a_{22} - a_{12}a_{21}.$$

→ the difference between the product of the diagonal elements and the product of the off-diagonal elements (a determinant of order 2).

- ▶ Example:

$$\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}, \quad |\mathbf{A}| = 3 \cdot 1 - 2 \cdot 1 = 1.$$

Matrix Operations: inverse

- ▶ The **inverse** of a square matrix $\mathbf{A}$, denoted $\mathbf{A}^{-1}$, is defined by $\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$

→ as in scalar algebra, where $\alpha\alpha^{-1} = 1$ if $\alpha \neq 0$).

- ▶ A matrix $\mathbf{A}$ is **invertible** iff:
 - it is square ($n \times n$) — *necessary condition*;
 - it is nonsingular, i.e. $|\mathbf{A}| \neq 0$ — *sufficient condition*.
- ▶ Properties of inverses:
 - $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$ (involution);
 - $(\mathbf{A}^{-1})' = (\mathbf{A}')^{-1}$;
 - $(\alpha\mathbf{A})^{-1} = \alpha^{-1}\mathbf{A}^{-1}$ for $\alpha \neq 0$.

Case of a 2×2 matrix

2×2 matrix $\mathbf{A}$:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix};$$

Determinant:

$$|\mathbf{A}| = a_{11}a_{22} - a_{12}a_{21}.$$

Inverse matrix:

$$\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}.$$

Extra slides – Computation of inverse matrix

Matrix Operations: determinant (*)

$n \times n$ **matrix** — **Laplace (cofactor) expansion**:

- ▶ $|\mathbf{A}|$ is a weighted sum of the determinants of n submatrices of $\mathbf{A}$, each of size $(n-1) \times (n-1)$;
- ▶ **minor** of a_{ij} , denoted M_{ij} : the subdeterminant obtained by deleting row i and column j of $|\mathbf{A}|$;
- ▶ **cofactor** C_{ij} : the minor with a sign attached, $C_{ij} = (-1)^{i+j} M_{ij}$;
- ▶ determinant of $\mathbf{A}$:

$$\begin{aligned} |\mathbf{A}| &= \sum_{j=1}^n a_{ij} C_{ij} \quad (\text{expansion by row } i) \\ &= \sum_{i=1}^n a_{ij} C_{ij} \quad (\text{expansion by column } j). \end{aligned}$$



Pierre-Simon Laplace
1749–1827

Matrix Operations: inverse (*) — using the **adjugate**

- General formula:

$$\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \text{adj}(\mathbf{A}),$$

where $\text{adj}(\mathbf{A})$ (the **adjoint** or **adjugate** of $\mathbf{A}$) is the transpose of the cofactor matrix $\mathbf{C}^+ = (C_{ij})$:

$$\text{adj}(\mathbf{A}) = (\mathbf{C}^+)^T = \begin{pmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{pmatrix}.$$

Matrix Operations: inverse (*) — worked example (i)

$$\mathbf{A} = \begin{pmatrix} 1 & 3 & 3 \\ 1 & 3 & 4 \\ 1 & 4 & 3 \end{pmatrix}, \quad \mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} (\mathbf{C}^+)^T.$$

Determinant (expanding along row 1):

$$|\mathbf{A}| = 1(3 \cdot 3 - 4 \cdot 4) - 3(1 \cdot 3 - 4 \cdot 1) + 3(1 \cdot 4 - 3 \cdot 1) = -7 + 3 + 3 = -1.$$

Cofactor matrix $\mathbf{C}^+ = (C_{ij})$:

$$\begin{aligned} C_{11} &= + \begin{vmatrix} 3 & 4 \\ 4 & 3 \end{vmatrix} = -7 & C_{12} &= - \begin{vmatrix} 1 & 4 \\ 1 & 3 \end{vmatrix} = 1 & C_{13} &= + \begin{vmatrix} 1 & 3 \\ 1 & 4 \end{vmatrix} = 1 \\ C_{21} &= - \begin{vmatrix} 3 & 3 \\ 4 & 3 \end{vmatrix} = 3 & C_{22} &= + \begin{vmatrix} 1 & 3 \\ 1 & 3 \end{vmatrix} = 0 & C_{23} &= - \begin{vmatrix} 1 & 3 \\ 1 & 4 \end{vmatrix} = -1 \\ C_{31} &= + \begin{vmatrix} 3 & 3 \\ 3 & 4 \end{vmatrix} = 3 & C_{32} &= - \begin{vmatrix} 1 & 3 \\ 1 & 4 \end{vmatrix} = -1 & C_{33} &= + \begin{vmatrix} 1 & 3 \\ 1 & 3 \end{vmatrix} = 0 \end{aligned}$$

Matrix Operations: inverse (*) — worked example (ii)

Cofactor matrix and its transpose (the adjugate):

$$\mathbf{C}^+ = \begin{pmatrix} -7 & 1 & 1 \\ 3 & 0 & -1 \\ 3 & -1 & 0 \end{pmatrix}, \quad (\mathbf{C}^+)^T = \begin{pmatrix} -7 & 3 & 3 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{pmatrix}.$$

Hence the inverse (with $|\mathbf{A}| = -1$):

$$\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} (\mathbf{C}^+)^T = \frac{1}{-1} \begin{pmatrix} -7 & 3 & 3 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{pmatrix} = \begin{pmatrix} 7 & -3 & -3 \\ -1 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix}.$$

Check: $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$.